1.1 Write transfers

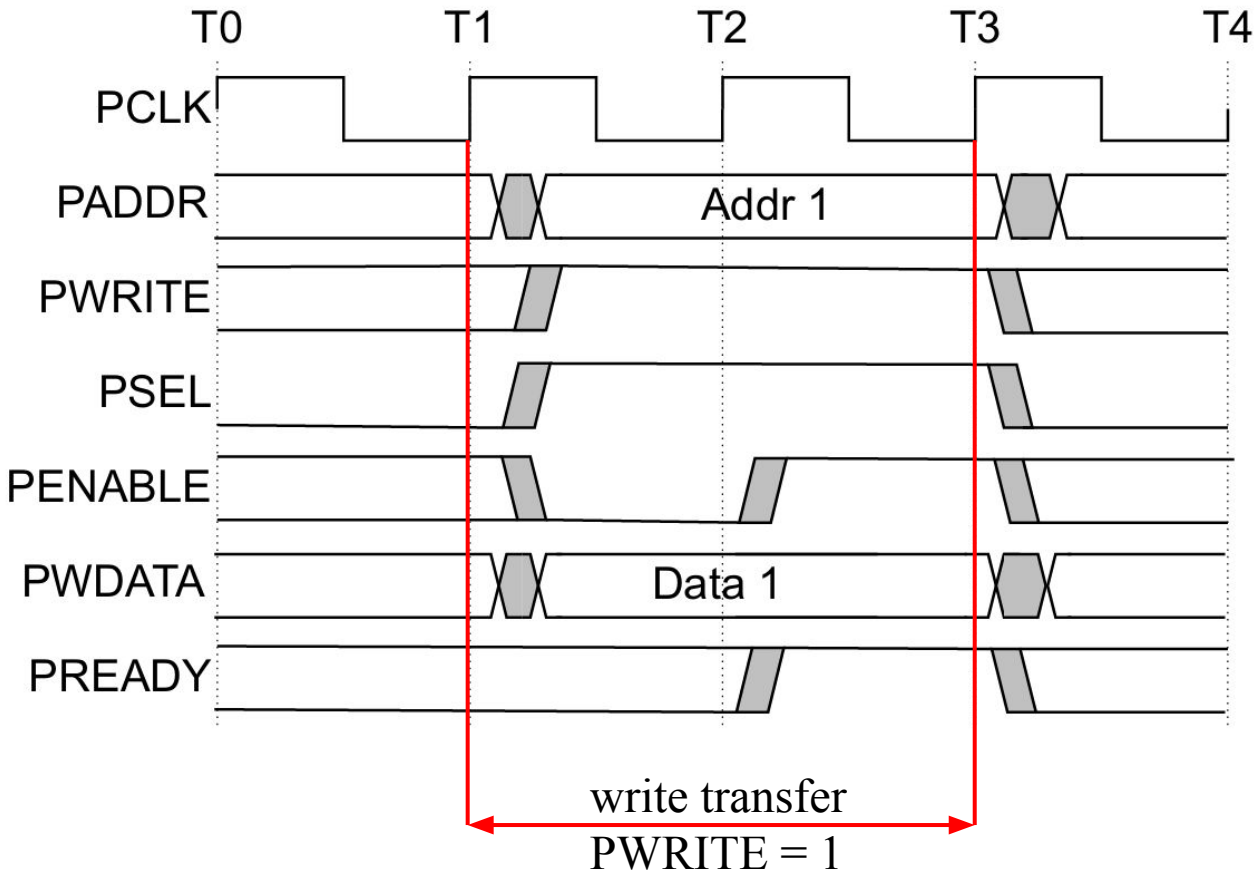
1.1.1 With no wait state

Setup Phase: T1-T2

- The select signal – **PSEL**, is asserted.
- **PADDR, PWRITE, PWDATA** must be valid.
- Exists in 1 clock.

Access phase: T2-T3

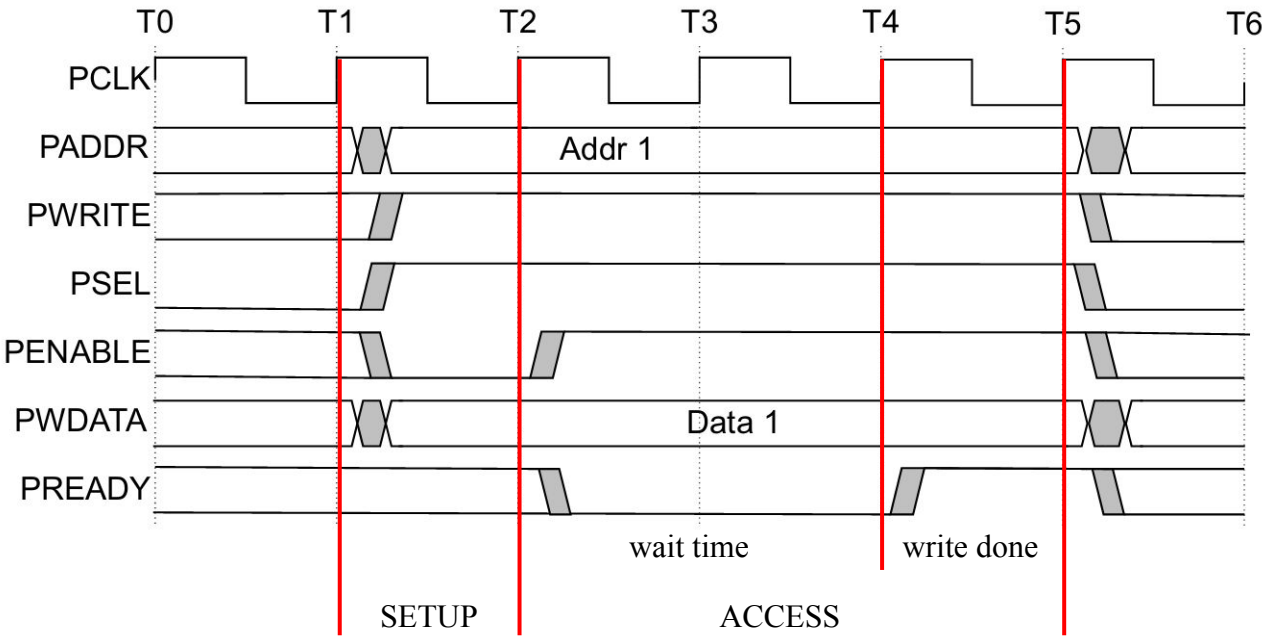
- **PENABLE** is asserted.
- Complete at T3, **PADDR, PWDATA** and any other control signals must be stable until transfer complete.



transfer must use 2 clock

1.1 Write transfers

1.1.2 With wait state

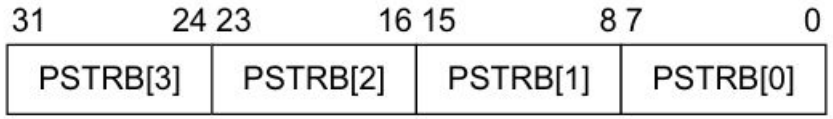


ACCESS PHASE:

- When **PENABLE=1, PREADY=0**, remain:
- PSEL : select signal
- PADDR : address signal
- PWRITE : direction signal
- PWDATA : write data signal
- PSTRB : write strobe signal
- PPROT : protection type signal
- PAUSER : user request attribute
- PWUSER : user write data attribute

1.1 Write transfers

1.1.1 With strobe



PSTRB[3:0]

1 bit strobe ⇔ 8 bits of write data bus

PSTRB[n] corresponds to **PWDATA[(8n+7):(8n)] (*)**

Note:

- Write transfer: each **PSTRB** corresponds to 1 byte of the write data bus.

  PSTRB[n] high ⇔ byte valid.
- Read transfer: the Requester must drive all bits of **PSTRB** LOW

Ex: data: 32bit: 12345678

PSTRB[3:0]=0001 => PWDATA[7:0]

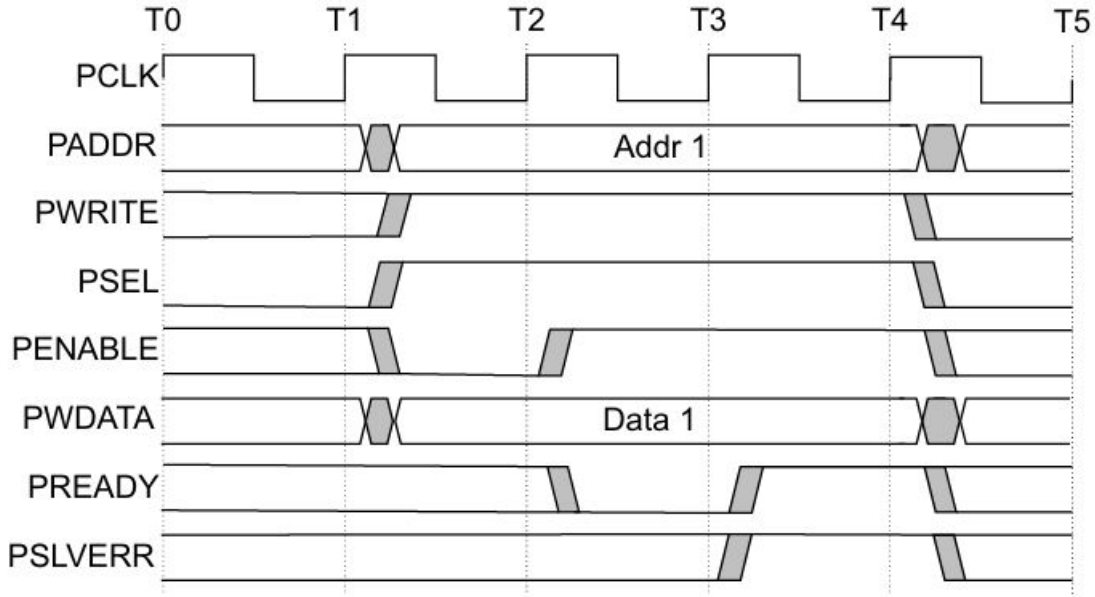
1.2 Error response

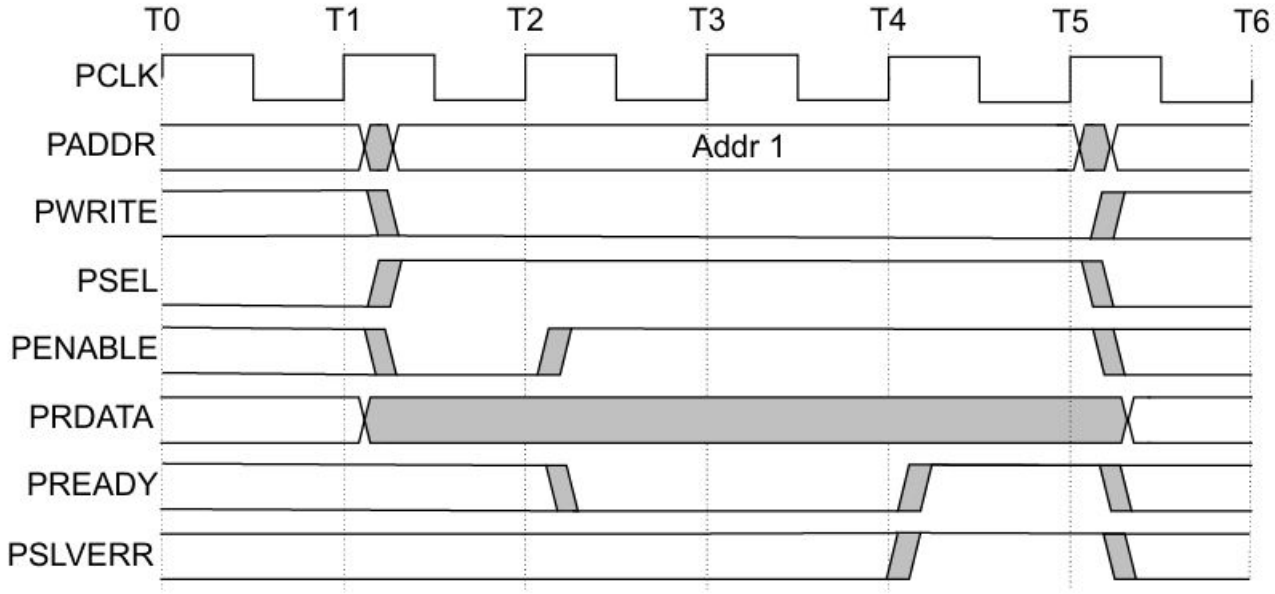
PSLVERR will be checked when completed transfer

(PREADY=1 – ACCESS PHASE).

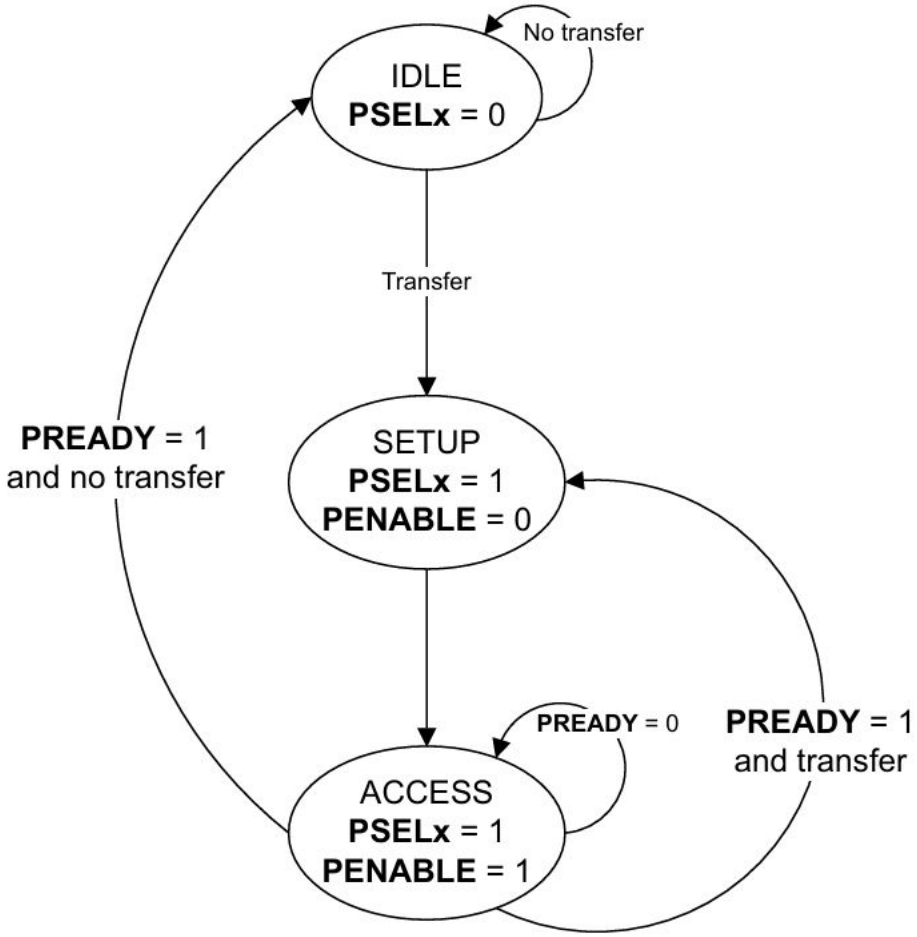
Can be used to indicate an error condition on an APB transfer.

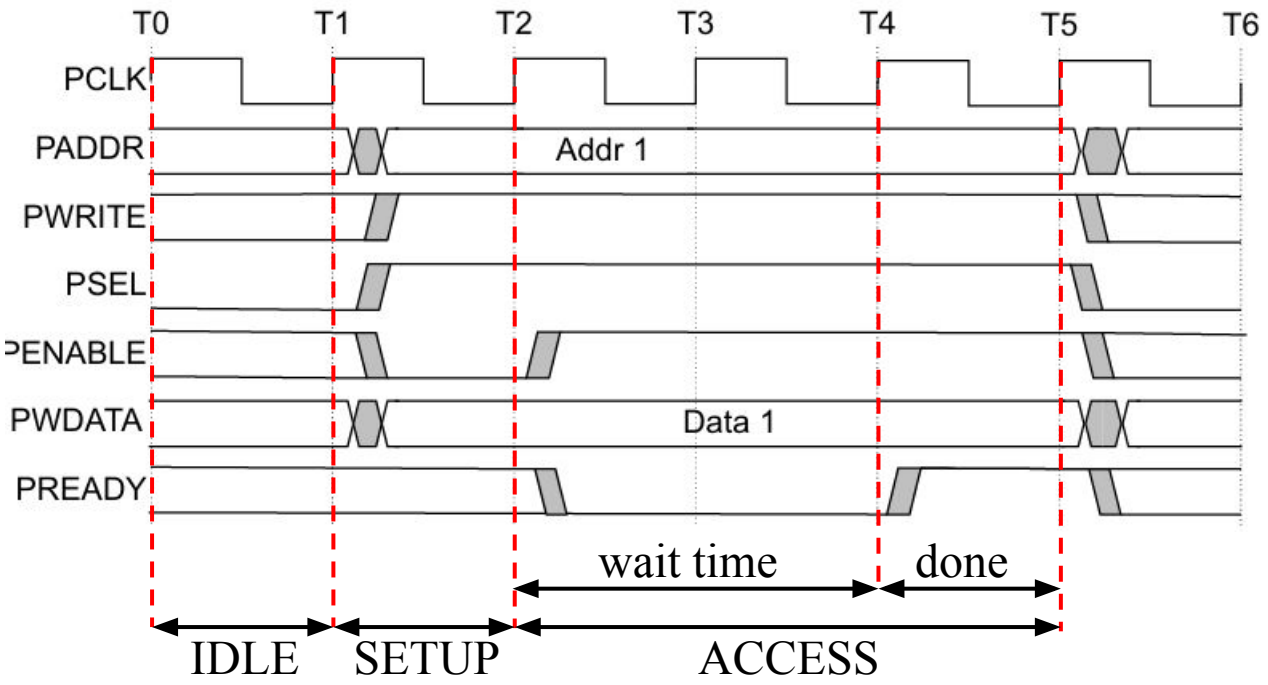
Recommend, not required, PSLVERR is LOW when no transfer.





FSM	APB PROTOCOL





IDLE: Default state – no transfer.

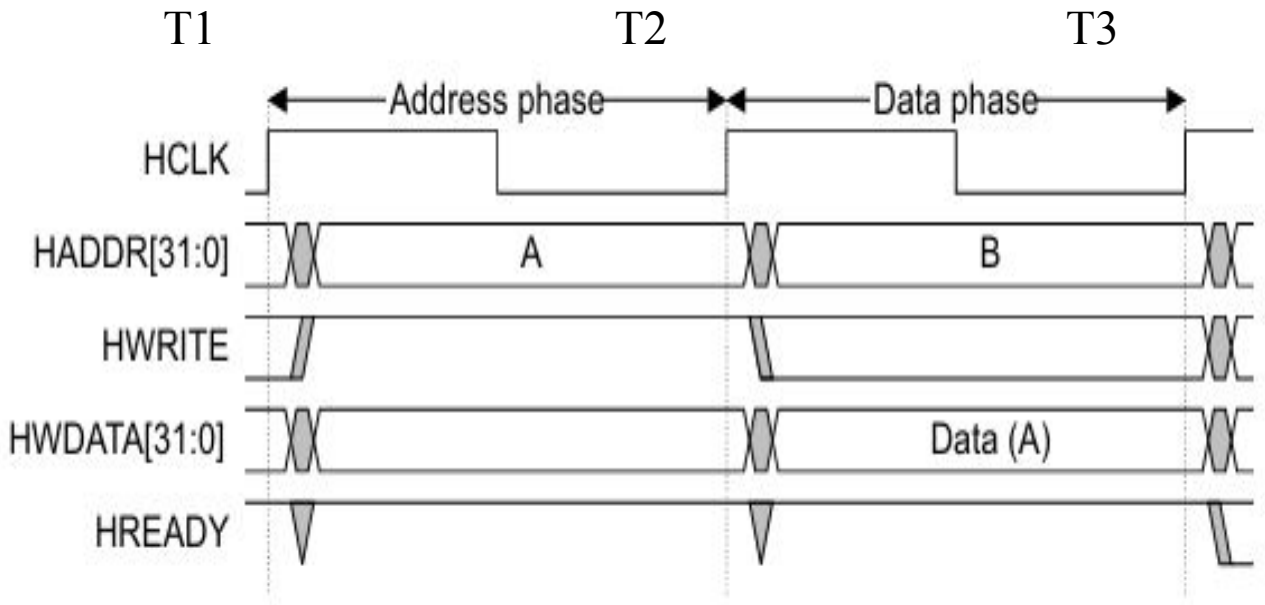
SETUP: Set up all signal for transfer without PENABLE. Exists in 1 clock.

ACCESS: wait time indicates time PREADY not asserted.

The following signals must not change in the transition between SETUP and ACCESS.

1.1 Write transfers

1.1.1 With no wait state



Address Phase: T1-T2

- **HSEL, HTRANS, HADDR, HWRITE, HREADY** must be valid in this phase.
- When write: **HWRITE = 1**.

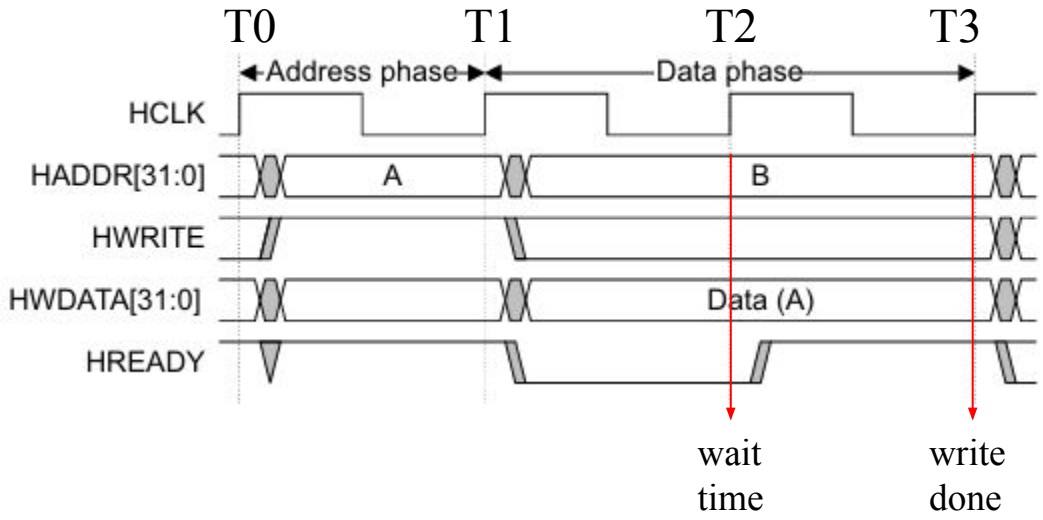
Data phase: T2-T3

- **HWDATA, HRDATA, HREADY, HRESP** must be valid in this phase.
- Complete at T3 when HREADY = 1.

Each phase will end and move on to the next phase when HREADY = 1.

1.1 Write transfers

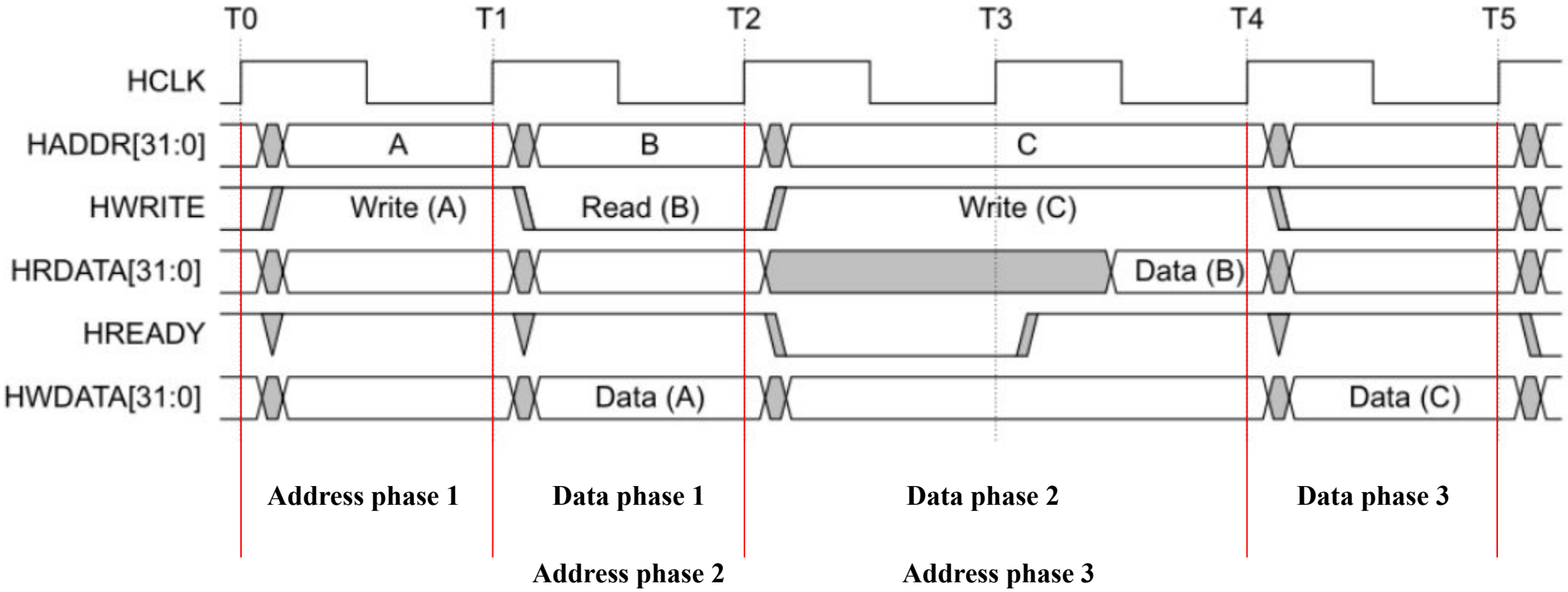
1.1.2 With wait state



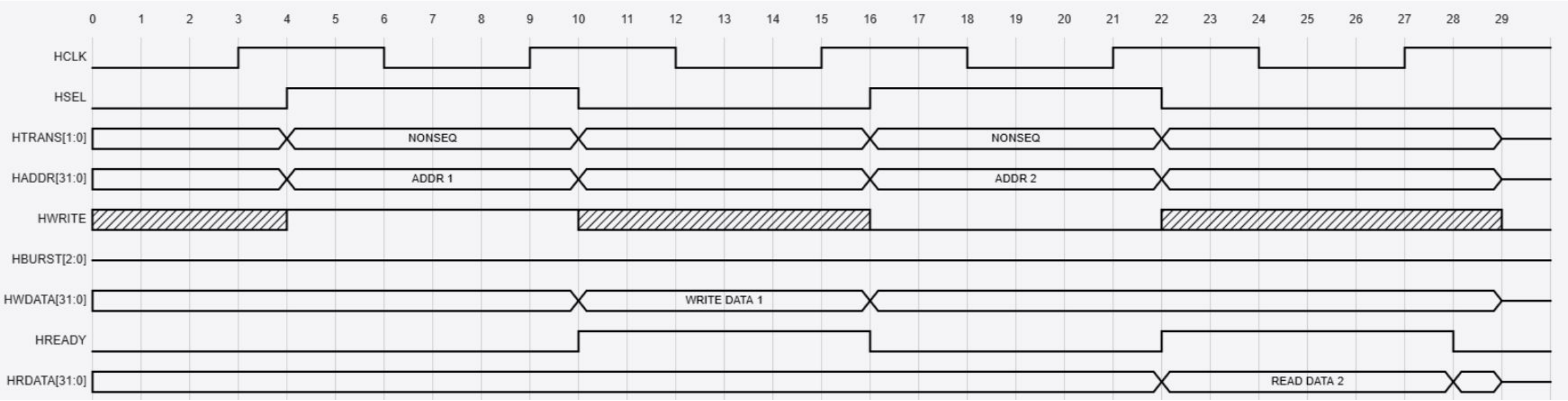
DATA PHASE:

- When HREADY = 0, the data can’t write so we have the wait time (T1 - T2).
- Wait time only exists when HREADY = 0.

1.1 Multiple transfers



HSEL	AHB PROTOCOL



HTRANS[1:0]	Type
0b00	IDLE
0b01	BUSY
0b10	NONSEQ
0b11	SEQ

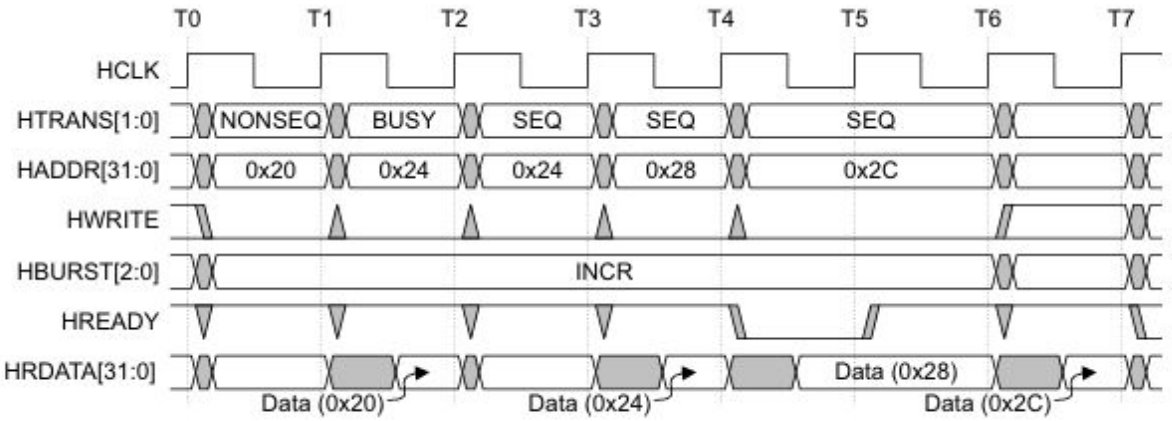


Figure 3-6 Transfer type examples

In Figure 3-6:

T0-T1 The 4-beat read starts with a NONSEQ transfer.

T1-T2 The Manager is unable to perform the second beat and inserts a BUSY transfer to delay the start of the second beat.

The Subordinate provides the read data for the first beat.

T2-T3 The Manager is now ready to start the second beat, so a SEQ transfer is signaled. The Manager ignores any data that the Subordinate provides on the read data bus.

T3-T4 The Manager performs the third beat.

The Subordinate provides the read data for the second beat.

T4-T5 The Manager performs the last beat.

The Subordinate is unable to complete the transfer and uses **HREADYOUT** to insert a single wait state.

T5-T6 The Subordinate provides the read data for the third beat.

T6-T7 The Subordinate provides the read data for the last beat.

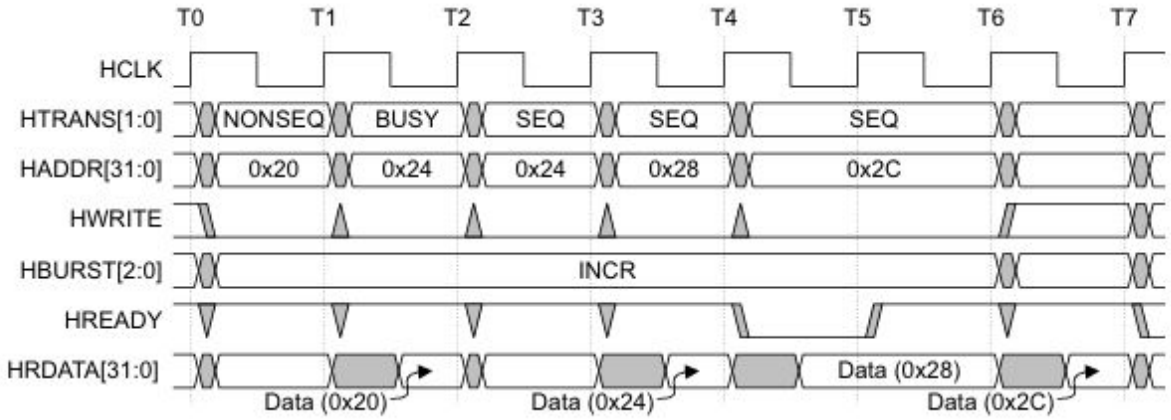


Figure 3-6 Transfer type examples

In Figure 3-6:

T0-T1 The 4-beat read starts with a NONSEQ transfer.

T1-T2 The Manager is unable to perform the second beat and inserts a BUSY transfer to delay the start of the second beat.

The Subordinate provides the read data for the first beat.

T2-T3 The Manager is now ready to start the second beat, so a SEQ transfer is signaled. The Manager ignores any data that the Subordinate provides on the read data bus.

T3-T4 The Manager performs the third beat.

The Subordinate provides the read data for the second beat.

T4-T5 The Manager performs the last beat.

The Subordinate is unable to complete the transfer and uses **HREADYOUT** to insert a single wait state.

T5-T6 The Subordinate provides the read data for the third beat.

T6-T7 The Subordinate provides the read data for the last beat.

HBURST[2:0]	Type	Description
0b000	SINGLE	Single transfer burst
0b001	INCR	Incrementing burst of undefined length
0b010	WRAP4	4-beat wrapping burst
0b011	INCR4	4-beat incrementing burst
0b100	WRAP8	8-beat wrapping burst
0b101	INCR8	8-beat incrementing burst
0b110	WRAP16	16-beat wrapping burst
0b111	INCR16	16-beat incrementing burst

Boundary = BEAT x HSIZE

0=0000
4=0100
8=1000
C=11100

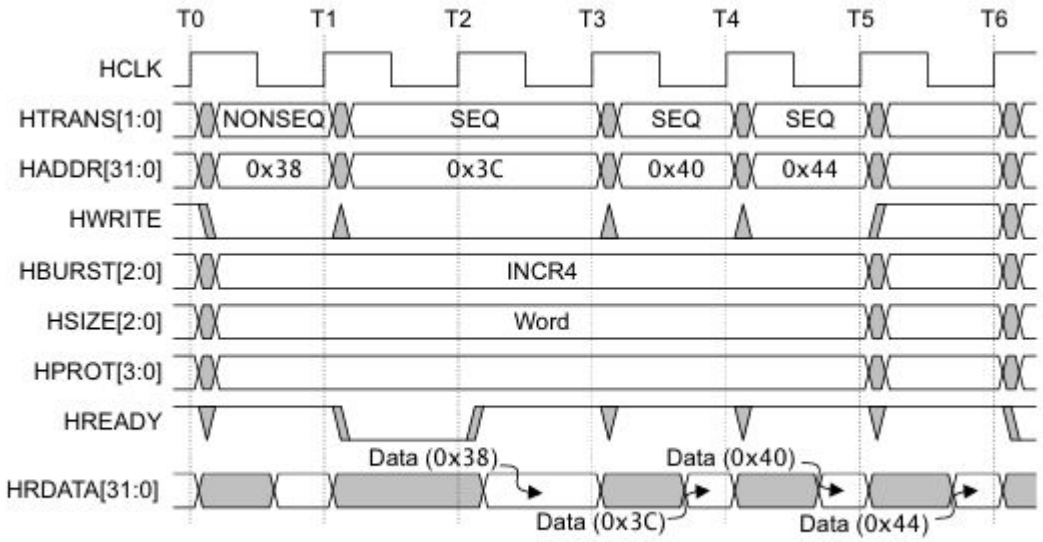


Figure 3-9 Four-beat incrementing burst

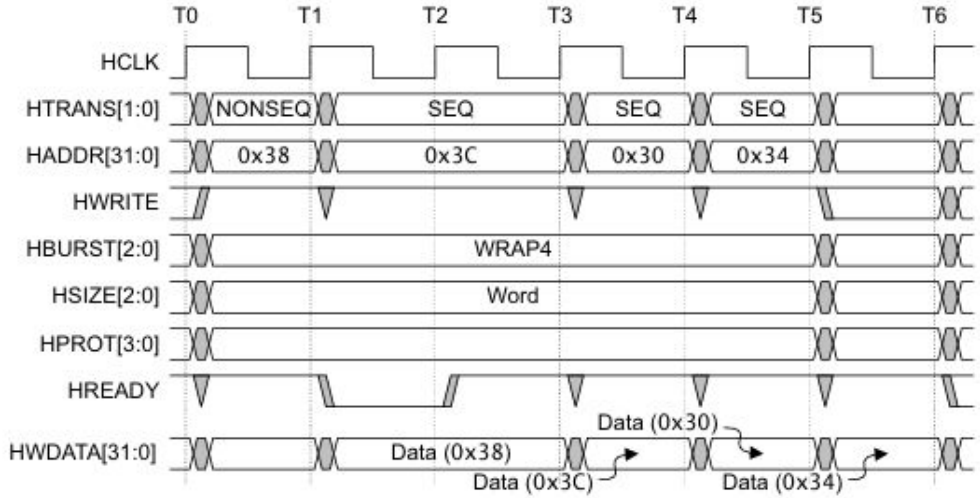
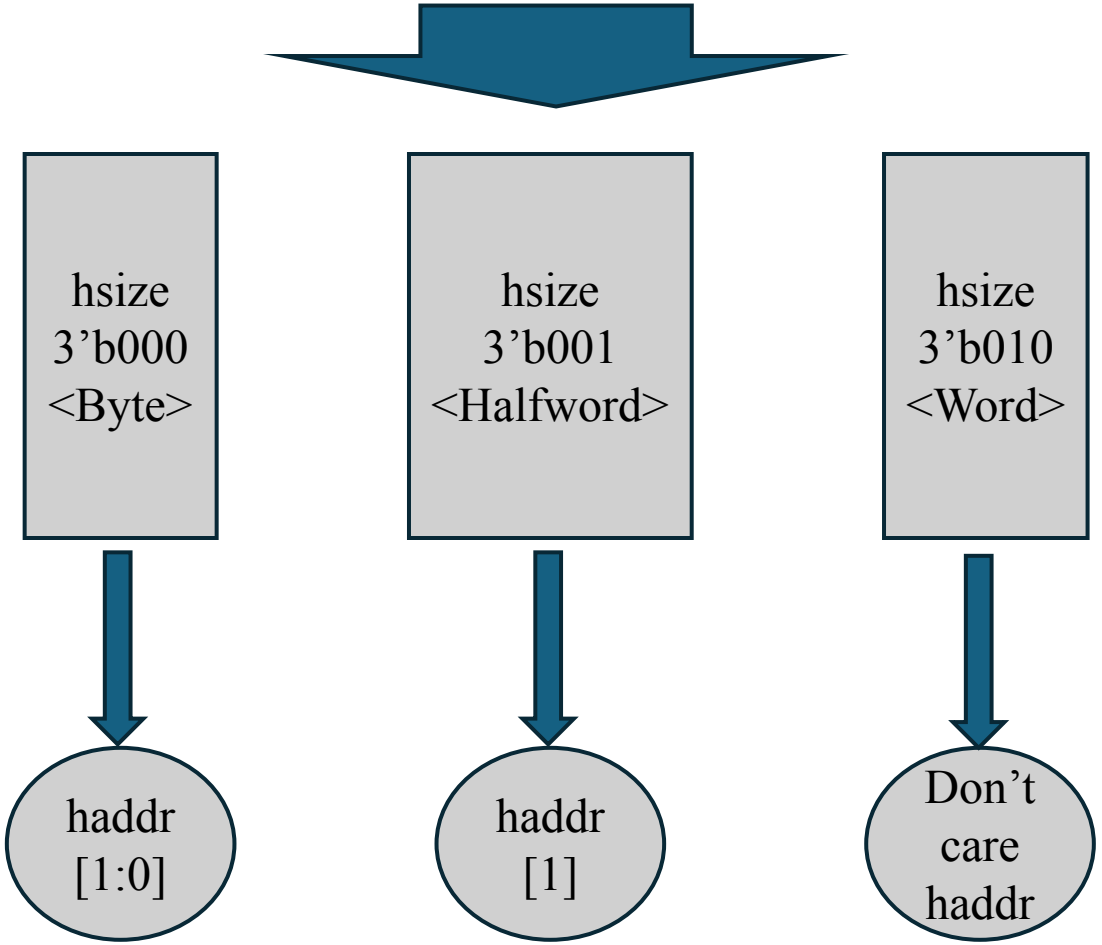


Figure 3-8 Four-beat wrapping burst

hsize is used to identify valid bytes to use in hwdata



Little-endian:

- Indicates the location of the byte on the data bus.
- Determined by Byte_Lane

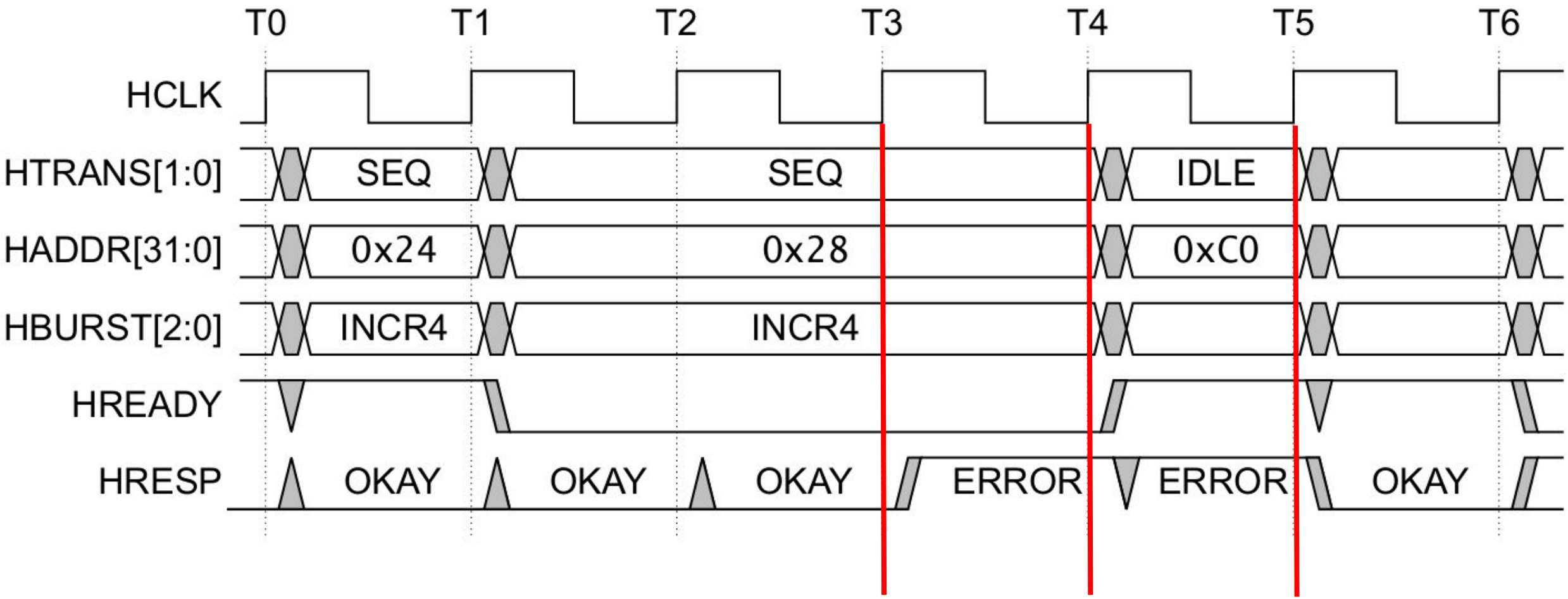
Data is transferred on

$$DATA[(8 \times Byte_Lane) + 7 : (8 \times Byte_Lane)]$$

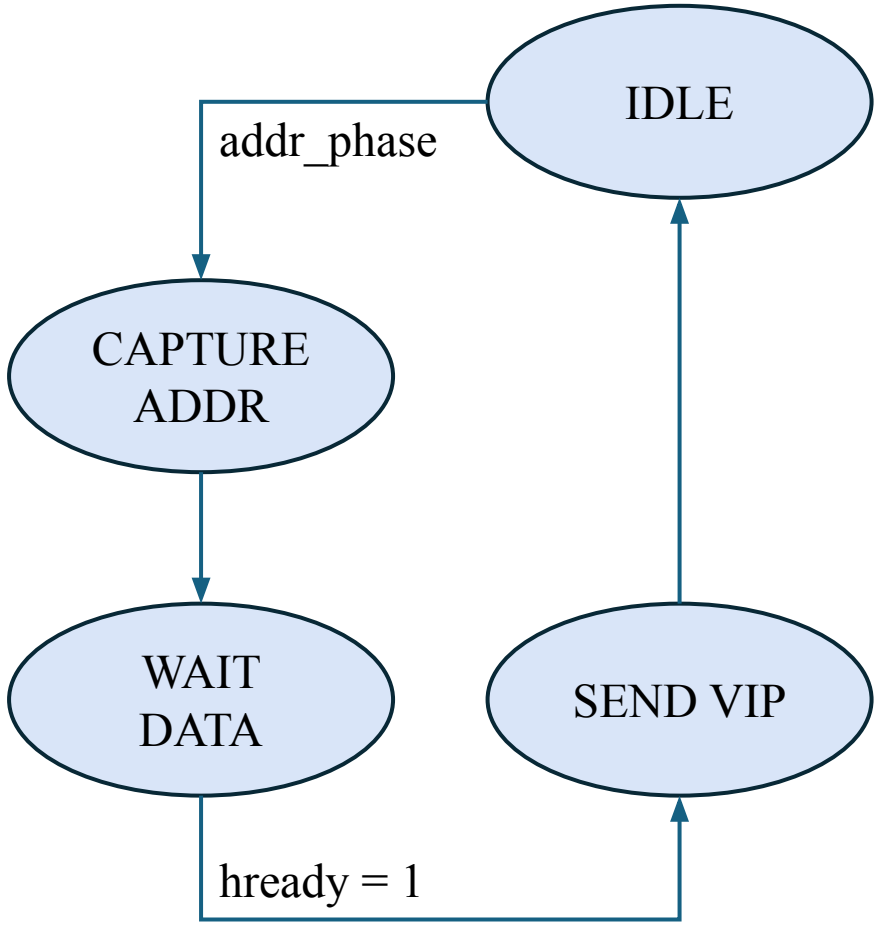
$$Byte_Lane = Address - \left(\left\lfloor \frac{Address}{Data_Bus_Bytes} \right\rfloor \times Data_Bus_Bytes \right)$$

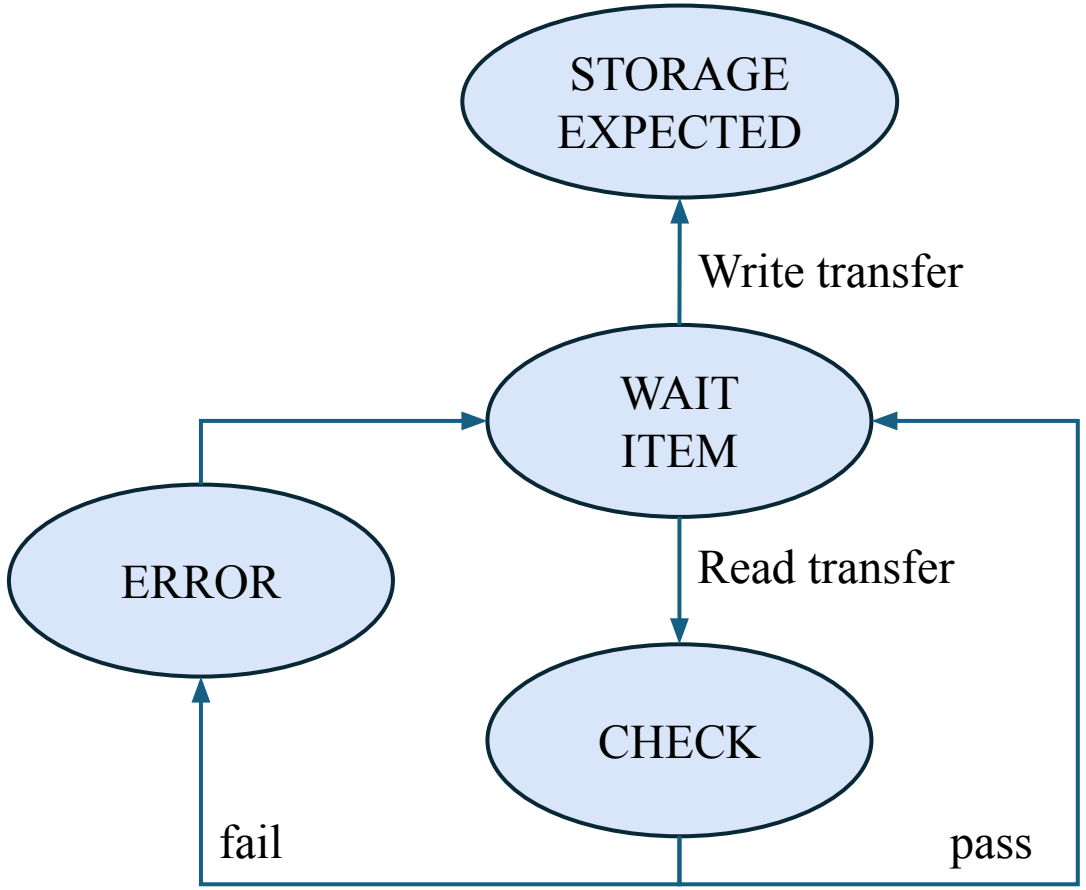
address	0x(n+3)	0x(n+2)	0x(n+1)	0x(n)
data	[31:24]	[23:16]	[15:8]	[7:0]
HSIZE	Byte 3	Byte 2	Byte 1	Byte 0
	Halfword 1		Halfword 0	
	Word			

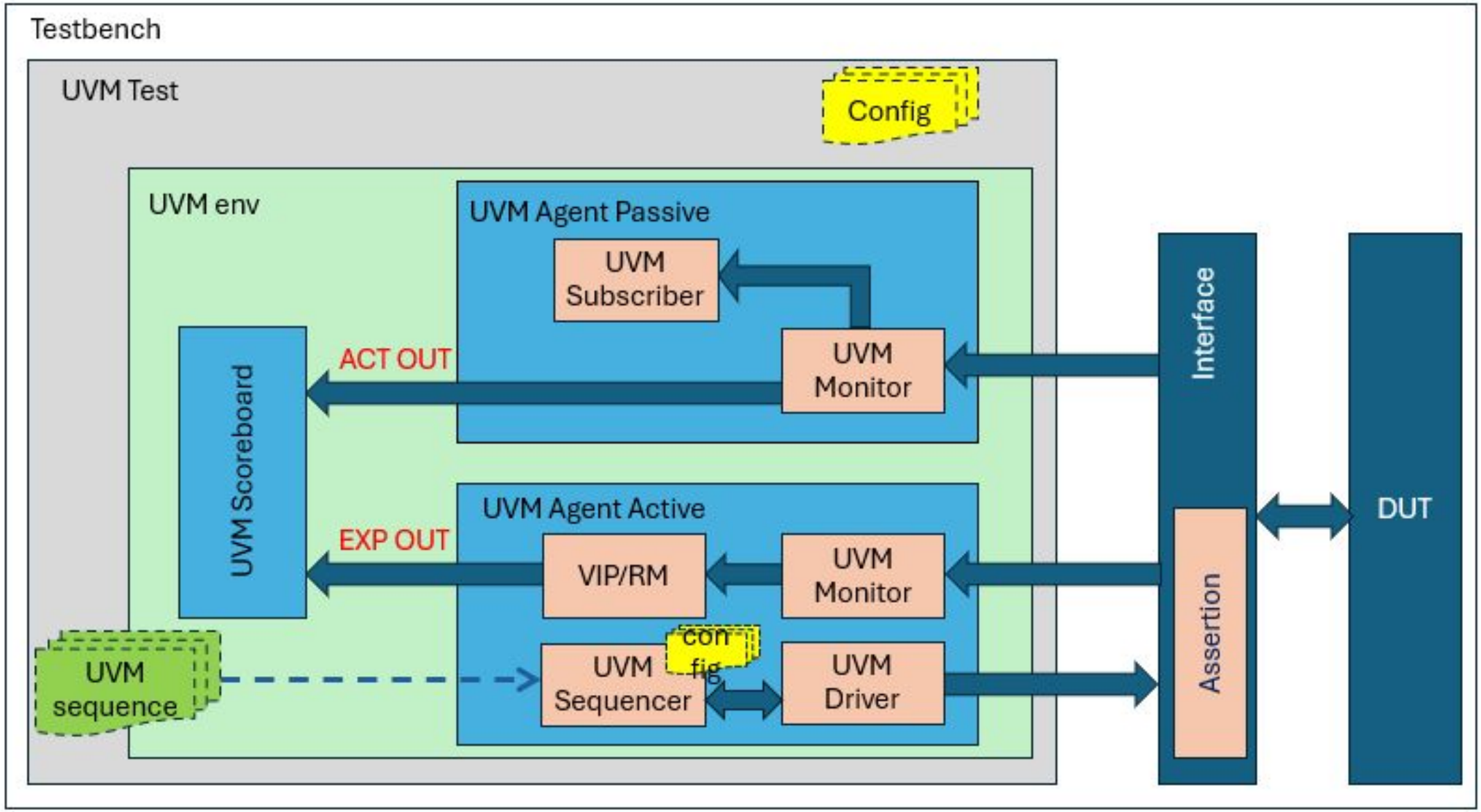
HRESP	AHB PROTOCOL



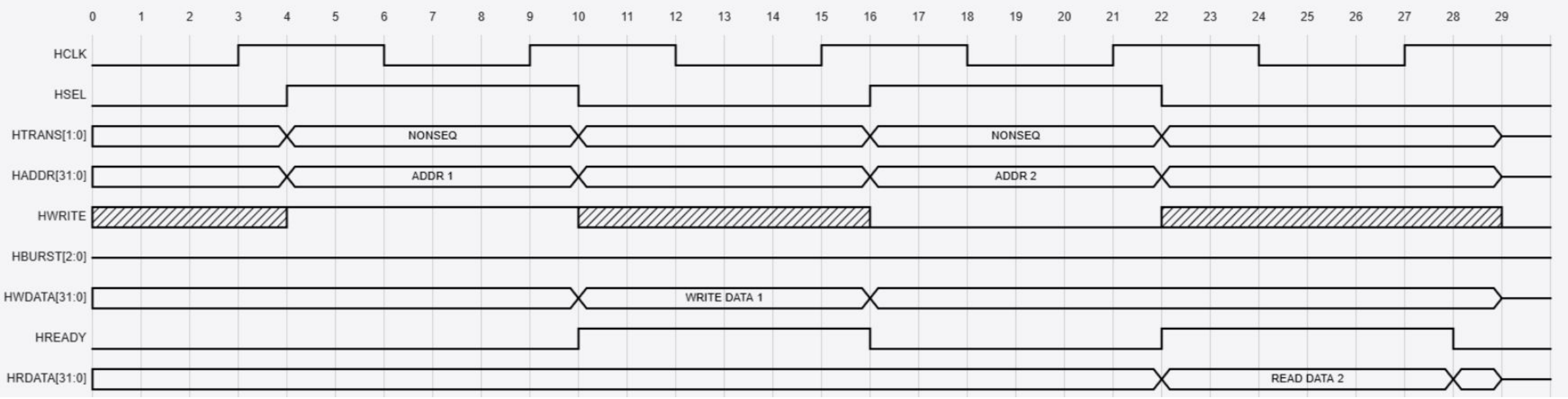
FSM


IDLE
addr_phase
CAPTURE
ADDR
WAIT
DATA
SEND VIP
hready = 1


STORAGE
EXPECTED
Write transfer
WAIT
ITEM
ERROR
Read transfer
CHECK
fail
pass


Testbench
UVM Test
Config
UVM env
UVM Agent Passive
UVM Subscriber
UVM Monitor
ACT OUT
UVM Scoreboard
UVM Agent Active
EXP OUT
VIP/RM
UVM Monitor
config
UVM Sequencer
UVM Driver
UVM sequence
Interface
Assertion
DUT

TIME DRV, MONITOR



Basic transfer – Single transfer:

- DRV : (clk_setup & clk_hold)
  - write transfer (hsel = 1, htrans = 2'b10, hsize, haddr, hwdata, hwrite = 1)
  - read transfer (hsel = 1, htrans = 2'b10, hsize, haddr, hrdata ,hwrite = 0)
- MON :
  - mon_active : hsel=1, htrans=2'b10, hwrite=0(actual)/1(expected)
    - create queue, store addr_phase, next clock, check transfer to write VIP
  - mon_passive : hsel=1, htrans=2'b10, hwrite=0